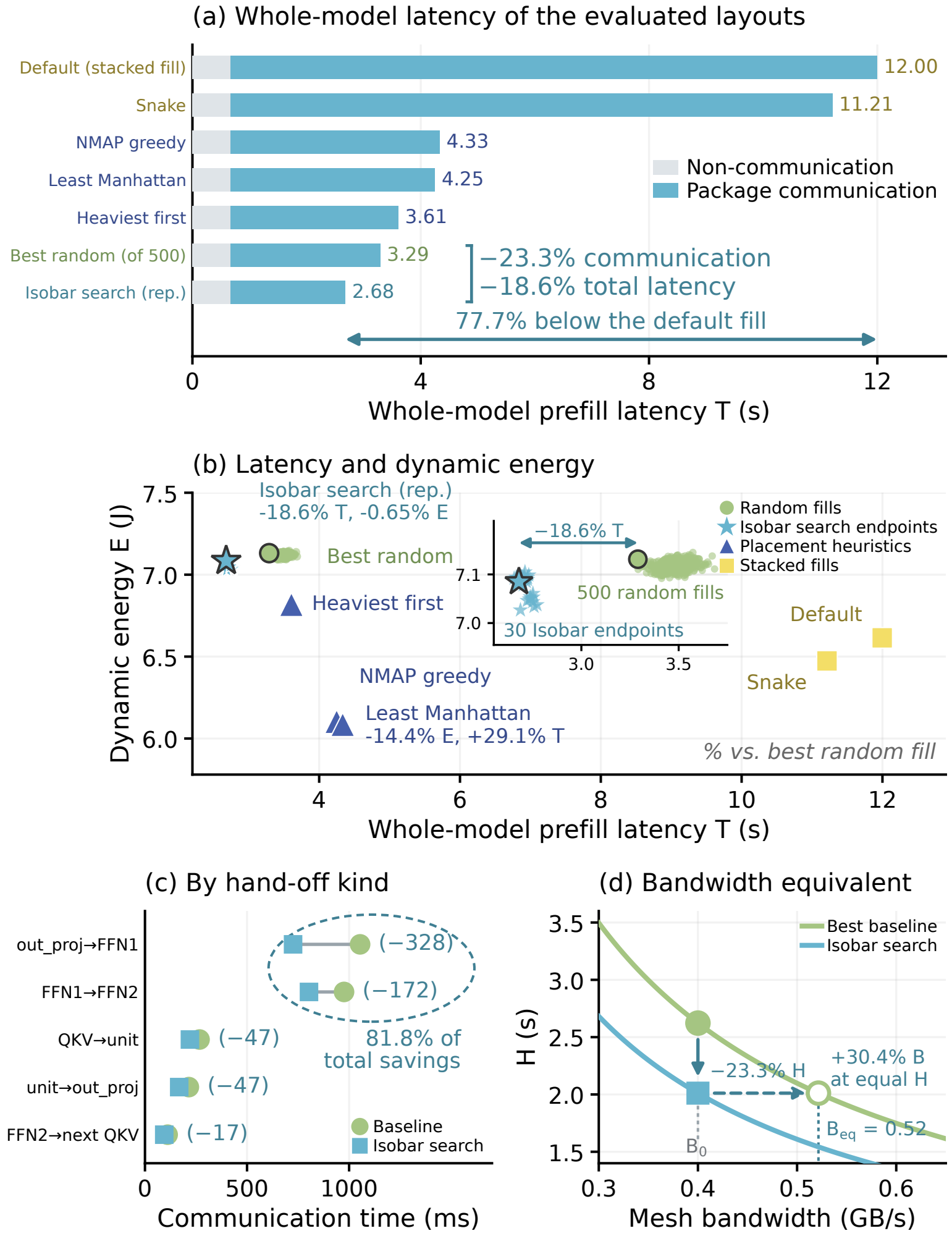
(a) Whole-model latency of the evaluated layouts
Default (stacked fill)
Snake
NMAP greedy
Least Manhattan
Heaviest first
Best random (of 500)
Isobar search (rep.)
12.00
11.21
4.33
4.25
3.61
3.29
2.68
Non-communication
Package communication
−23.3% communication
−18.6% total latency
77.7% below the default fill
0
4
8
12
Whole-model prefill latency T (s)
(b) Latency and dynamic energy
Isobar search (rep.)
-18.6% T, -0.65% E
Best random
Heaviest first
NMAP greedy
Least Manhattan
-14.4% E, +29.1% T
Default
Snake
% vs. best random fill
−18.6% T
500 random fills
30 Isobar endpoints
Random fills
Isobar search endpoints
Placement heuristics
Stacked fills
Dynamic energy E (J)
7.5
7.0
6.5
6.0
4
6
8
10
12
Whole-model prefill latency T (s)
(c) By hand-off kind
out_proj→FFN1
FFN1→FFN2
QKV→unit
unit→out_proj
FFN2→next QKV
(−328)
(−172)
(−47)
(−47)
(−17)
81.8% of
total savings
Baseline
Isobar search
0
500
1000
Communication time (ms)
(d) Bandwidth equivalent
Best baseline
Isobar search
H (s)
3.5
3.0
2.5
2.0
1.5
−23.3% H
+30.4% B
at equal H
B0
Beq = 0.52
0.3
0.4
0.5
0.6
Mesh bandwidth (GB/s)